

```
ubuntu@ip-172-31-10-53:~$ #Lab59. Monitoring with top and htop_lab
ubuntu@ip-172-31-10-53:~$ #OBJECTIVES:Understand the purpose and importance of system
monitoring tools top and htop.
ubuntu@ip-172-31-10-53:~$
ubuntu@ip-172-31-10-53:~$ #Task1:Installation of top&htop
ubuntu@ip-172-31-10-53:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2049
kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [360 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components
[177 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata
[17.1 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages
[1694 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [330
kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64
Components [386 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Icons (64x64) [373
kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f
Metadata [34.5 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages
[3278 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en
[760 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64
Components [212 B]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f
Metadata [480 B]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages
[44.7 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en
[11.3 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64
Components [940 B]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f
Metadata [656 B]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages
[40.6 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9172
B]
```

Get:24 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/main amd64 Components [5760 B]
Get:25 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/main Icons (48x48) [9535 B]
Get:26 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/main Icons (64x64) [11.3 kB]
Get:27 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/main amd64 c-n-f Metadata [368 B]
Get:28 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/universe amd64 Packages [31.0 kB]
Get:29 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/universe Translation-en [18.6 kB]
Get:30 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/universe amd64 Components [10.5 kB]
Get:31 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/universe Icons (48x48) [23.3 kB]
Get:32 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/universe Icons (64x64) [32.3 kB]
Get:33 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/universe amd64 c-n-f Metadata [1484 B]
Get:34 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/restricted amd64 Components [212 B]
Get:35 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/multiverse amd64 Packages [748 B]
Get:36 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/multiverse Translation-en [340 B]
Get:37 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-backports/multiverse amd64 Components [212 B]
Get:38 <http://security.ubuntu.com/ubuntu> noble-security/main amd64 Packages [1704 kB]
Get:39 <http://security.ubuntu.com/ubuntu> noble-security/main Translation-en [268 kB]
Get:40 <http://security.ubuntu.com/ubuntu> noble-security/main amd64 Components [42.5 kB]
Get:41 <http://security.ubuntu.com/ubuntu> noble-security/main Icons (48x48) [14.7 kB]
Get:42 <http://security.ubuntu.com/ubuntu> noble-security/main Icons (64x64) [21.1 kB]
Get:43 <http://security.ubuntu.com/ubuntu> noble-security/main amd64 c-n-f Metadata [11.0 kB]
Get:44 <http://security.ubuntu.com/ubuntu> noble-security/universe amd64 Packages [1192 kB]
Get:45 <http://security.ubuntu.com/ubuntu> noble-security/universe Translation-en [230 kB]
Get:46 <http://security.ubuntu.com/ubuntu> noble-security/universe amd64 Components [74.2 kB]
Get:47 <http://security.ubuntu.com/ubuntu> noble-security/universe amd64 c-n-f Metadata [23.1 kB]
Get:48 <http://security.ubuntu.com/ubuntu> noble-security/restricted amd64 Packages [3006 kB]
Get:49 <http://security.ubuntu.com/ubuntu> noble-security/restricted Translation-en [698 kB]
Get:50 <http://security.ubuntu.com/ubuntu> noble-security/restricted amd64 Components [212 B]
Get:51 <http://security.ubuntu.com/ubuntu> noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:52 <http://security.ubuntu.com/ubuntu> noble-security/multiverse amd64 Packages [39.3 kB]
Get:53 <http://security.ubuntu.com/ubuntu> noble-security/multiverse Translation-en [9000 B]
Get:54 <http://security.ubuntu.com/ubuntu> noble-security/multiverse amd64 Components [208 B]
Get:55 <http://security.ubuntu.com/ubuntu> noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 17.4 MB in 3s (5312 kB/s)

```

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
374 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-10-53:~$ sudo apt install htop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
htop is already the newest version (3.3.0-4build1).
htop set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-aws-6.14-headers-6.14.0-1015 linux-aws-6.14-tools-6.14.0-1015 linux-headers-6.14.0-1015-
aws linux-image-6.14.0-1015-aws
  linux-modules-6.14.0-1015-aws linux-tools-6.14.0-1015-aws
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 374 not upgraded.
ubuntu@ip-172-31-10-53:~$
ubuntu@ip-172-31-10-53:~$ #Task2:Running htop&top
ubuntu@ip-172-31-10-53:~$ top
top - 05:38:14 up 5 min,  3 users,  load average: 0.32, 0.33, 0.16
Tasks: 217 total,  1 running, 216 sleeping,  0 stopped,  0 zombie
%Cpu(s):  3.3 us,  2.5 sy,  0.0 ni, 93.6 id,  0.0 wa,  0.0 hi,  0.0 si,  0.7 st
MiB Mem :  3837.0 total,  1413.3 free,  1258.1 used,  1427.7 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  2578.8 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2037	ubuntu	20	0	1552280	150220	102560	S	3.3	3.8	0:09.57	dcvagent
1521	ubuntu	9	-11	110052	10156	7296	S	2.3	0.3	0:04.93	pipewire
738	dcd	20	0	634460	49052	40416	S	1.3	1.2	0:03.59	dcdserver
1524	ubuntu	9	-11	118216	17592	10120	S	1.3	0.4	0:03.94	pipewire-pulse
2192	cdagent	20	0	1412048	108016	84156	S	0.7	2.7	0:01.18	amazon-cloudwat
5288	ubuntu	20	0	12348	5900	3680	R	0.7	0.2	0:00.06	top
1552	ubuntu	20	0	2352188	90212	60680	S	0.3	2.3	0:01.27	Xorg
2354	ubuntu	20	0	2798480	58348	43688	S	0.3	1.5	0:00.61	gjs
1	root	20	0	23296	14564	9624	S	0.0	0.4	0:02.93	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0-events
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
11	root	20	0	0	0	0	I	0.0	0.0	0:00.05	kworker/0:1-cgroup_destroy
12	root	20	0	0	0	0	I	0.0	0.0	0:00.16	kworker/u8:0-flush-259:0
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq

```

14 root      20   0   0   0   0 I  0.0  0.0  0:00.00 rcu_tasks_rude_kthread
15 root      20   0   0   0   0 I  0.0  0.0  0:00.00 rcu_tasks_trace_kthread
16 root      20   0   0   0   0 S  0.0  0.0  0:00.07 ksoftirqd/0

```

ubuntu@ip-172-31-10-53:~\$

ubuntu@ip-172-31-10-53:~\$ htop

ubuntu@ip-172-31-10-53:~\$ htop

ubuntu@ip-172-31-10-53:~\$ t

t: command not found

ubuntu@ip-172-31-10-53:~\$ top

top - 05:40:59 up 8 min, 3 users, load average: 0.12, 0.26, 0.17

Tasks: 209 total, 1 running, 208 sleeping, 0 stopped, 0 zombie

%Cpu(s): 2.3 us, 2.5 sy, 0.0 ni, 95.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 3837.0 total, 1412.5 free, 1257.0 used, 1431.0 buff/cache

MiB Swap: 0.0 total, 0.0 free, 0.0 used. 2580.0 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2037	ubuntu	20	0	1552280	150220	102560	S	3.0	3.8	0:15.29	dcbagent
1521	ubuntu	9	-11	110052	10156	7296	S	1.7	0.3	0:08.30	pipewire
1524	ubuntu	9	-11	118216	17592	10120	S	1.7	0.4	0:06.53	pipewire-pulse
738	dcb	20	0	634460	49068	40416	S	1.0	1.2	0:05.43	dcbserver
2192	cbagent	20	0	1412048	108044	84156	S	0.7	2.7	0:01.76	amazon-cloudwat
6463	ubuntu	20	0	12348	5888	3668	R	0.7	0.1	0:00.07	top
190	root	20	0	0	0	0	I	0.3	0.0	0:00.06	kworker/0:3-events
1552	ubuntu	20	0	2352188	90212	60680	S	0.3	2.3	0:01.66	Xorg
2823	ubuntu	20	0	7740	3604	3324	S	0.3	0.1	0:00.77	bash
1	root	20	0	23296	14564	9624	S	0.0	0.4	0:02.95	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
12	root	20	0	0	0	0	I	0.0	0.0	0:00.18	kworker/u8:0-flush-259:0
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
16	root	20	0	0	0	0	S	0.0	0.0	0:00.08	ksoftirqd/0
17	root	20	0	0	0	0	I	0.0	0.0	0:00.30	rcu_sched

ubuntu@ip-172-31-10-53:~\$ htop

ubuntu@ip-172-31-10-53:~\$ top -b -n 1 or ps aux --sort=-%cpu | head -n 20

top: unrecognized field name '%cpu'

ubuntu@ip-172-31-10-53:~\$ top -b -n 1 or ps aux --sort=%cpu | head -n 20

top: unrecognized field name '%cpu'

ubuntu@ip-172-31-10-53:~\$ top -b -n 1 or ps aux --sort=-%CPU | head -n 20

top: unknown option 'or'

```
ubuntu@ip-172-31-10-53:~$ top -b -n 1 | head -n 40
```

```
top - 05:43:47 up 11 min, 3 users, load average: 0.21, 0.20, 0.16
```

```
Tasks: 210 total, 1 running, 209 sleeping, 0 stopped, 0 zombie
```

```
%Cpu(s): 4.8 us, 4.8 sy, 0.0 ni, 90.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

```
MiB Mem : 3837.0 total, 1431.3 free, 1238.1 used, 1431.1 buff/cache
```

```
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 2598.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
17	root	20	0	0	0	0	I	10.0	0.0	0:00.35	rcu_sch+
1	root	20	0	23296	14564	9624	S	0.0	0.4	0:02.95	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_wo+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
12	root	20	0	0	0	0	I	0.0	0.0	0:00.19	kworker+
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tas+
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tas+
16	root	20	0	0	0	0	S	0.0	0.0	0:00.09	ksoftir+
18	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp+
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp+
20	root	rt	0	0	0	0	S	0.0	0.0	0:00.03	migrati+
21	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_in+
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
24	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_in+
25	root	rt	0	0	0	0	S	0.0	0.0	0:00.10	migrati+
26	root	20	0	0	0	0	S	0.0	0.0	0:00.09	ksoftir+
27	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker+
28	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
29	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmp+
30	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
31	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
32	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungta+
33	root	20	0	0	0	0	I	0.0	0.0	0:00.11	kworker+
34	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_rea+
35	root	20	0	0	0	0	I	0.0	0.0	0:00.12	kworker+

```
ubuntu@ip-172-31-10-53:~$
```

ubuntu@ip-172-31-10-53:~\$ #CONCLUSION:The top and htop commands are essential tools for monitoring system processes on Linux. While top is a standard utility available by default, htop offers advanced features such as an interactive user interface and easy navigability. Mastering these tools can significantly aid in system administration tasks, such as identifying high resource-utilization processes and making data-driven decisions to optimize system performance.

ubuntu@ip-172-31-10-53:~\$

ubuntu@ip-172-31-10-53:~\$ #LabDone